

Dynamic Soil-Structure Interaction : Stochastic simulations

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Abstract

This paper deals with several numerical techniques accounting for random excitation and random parameters classically occurring in earthquake engineering. It studies in sequence the influence on the structural response of the variability of the incident field using filtering theory, the soil variability using Monte-Carlo simulation, and at last the influence on the site response of the a random building distribution using periodic simulations. In each case practical examples are given.

1 Stochastic modeling in earthquake engineering

Soil-structure interaction has received a broad attention in the recent years due to the significant role it plays on the seismic response of massive structures such as nuclear power plants, arch dams or even smaller buildings resting on soft deposits. As a starting point of this study, a mathematical model representing this phenomena is given in section 2.

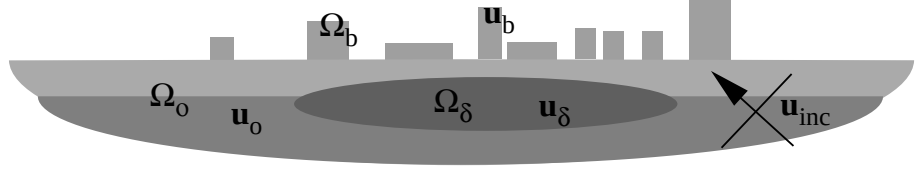
Earthquakes are by essence uncertain events not only regarding their occurrence and location -not addressed in this paper- but also during the event itself. Indeed recordings of strong motion arrays show a stochastic behaviour of the seismic field both in the time and space variables even over short distances. This variability justifies the use of stochastic incident field in our mathematical model which is solved in section 3 using the standard filtering theory.

The variability of the incident field is partly due to the variability of the soil itself. Then the structural dynamic response may be affected not only throughout this loading but also because of diffusion or localisation effects on these random heterogeneities of the soil. These two phenomena having contradictory consequences on the safety, the effect of the soil randomness is added in our model in section 4. For these models Monte-Carlo simulations will be used to find a constructive solution.

Coming back to seismic field, it has been observed that a large collection of buildings can significantly modify it. This phenomenon denoted the “site-city effect” is studied in section 5 assuming an infinite number of buildings showing an homogeneous random distribution in space. The corresponding model will be accounted for using large periodic models.

2 The generic deterministic model

Let Ω_o be the *reference* stratified half-space with known physical properties $(\lambda_o, \mu_o, \rho_o)$ (the Lam coefficients and the mass density), P_o being its planar free surface. Then let Ω_δ be the bounded soil *perturbations* characterized by the fluctuations of the physical properties $\mathbf{p}_\delta = (\lambda_\delta, \mu_\delta, \rho_\delta)$ and Ω_b the deterministic or undeterministic *structures* resting on Ω_o , Σ denoting the planar interface between Ω_b and Ω_o . The actual domain denoted Ω is the

FIG. 1. *The generic model*

superposition of these three sub-domains Ω_o , Ω_δ and Ω_b , and one assumes for simplicity that $\partial\Omega_\delta \cap P_o = \emptyset$. Moreover one uses the following notations : $\mathbf{u}_\beta$ and $\mathbf{v}_\beta$ denote displacement fields on the corresponding sub-domain Ω_β , $\boldsymbol{\epsilon}(\mathbf{v}_\beta)$ the associated strain tensor, $\boldsymbol{\sigma}_\beta(\mathbf{v}_\beta)$ the stress tensor computed with the elastic tensor $\mathbf{C}_\alpha$ derived from $\mathbf{p}_\delta$ and $\mathbf{Div}$ denotes the divergence. $(\mathbf{u}_\beta, \mathbf{v}_\beta)_\beta$ denotes the usual scalar product on sub-domain Ω_β , H_β denoting the Hilbert space of fields having a finite norm and V_β the subspace of H_β whose elements have all their derivatives in H_β .

The loading on Ω is an incident field $\mathbf{u}^{inc}$ satisfying the Navier equation in Ω_o . It can be replaced formally by the following dynamic loads $\mathbf{f}_s$ and the following jump condition along Σ at each circular frequency ω :

$$(1) \quad \mathbf{f}_{inc} = \mathbf{Div} \boldsymbol{\sigma}_\delta(\mathbf{u}^{inc}) + \rho_\delta \omega^2 \mathbf{u}^{inc} \quad , \quad [\mathbf{u}] = \mathbf{u}_o - \mathbf{u}_b = \mathbf{u}^{inc}$$

and one has to solve the following variational problem for the *diffracted field* $\mathbf{u}$:

$$(2) \quad (\mathbf{f}_s, \mathbf{v}_\delta)_\delta = k_o(\mathbf{u}_o, \mathbf{v}_o) + k_\delta(\mathbf{u}_\delta, \mathbf{v}_\delta) + k_b(\mathbf{u}_b, \mathbf{v}_b)$$

$$(3) \quad k_\beta(\mathbf{u}_\beta, \mathbf{v}_\beta) \stackrel{\text{def}}{=} (\boldsymbol{\sigma}_\beta(\mathbf{u}_\beta), \boldsymbol{\epsilon}(\mathbf{v}_\beta))_\beta - \omega^2 (\rho_\beta \mathbf{u}_\beta, \mathbf{v}_\beta)_\beta$$

$$(4) \quad (\mathbf{u}_o + \mathbf{u}^{inc})|_\Sigma = \mathbf{u}_b|_\Sigma \quad \mathbf{u}_o|_{\Omega_\delta} = \mathbf{u}_\delta \quad \mathbf{v}_o|_\Sigma = \mathbf{v}_b|_\Sigma \quad \mathbf{v}_o|_{\Omega_\delta} = \mathbf{v}_\delta$$

As long damping is accounted for, this problem has an unique solution in the proper subspace of $V_o \times V_\delta \times V_b$ satisfying the compatibility conditions (4). Moreover, when Σ and Ω_δ are bounded, this deterministic problem may be efficiently solved using sub-structuring techniques [3, 2] and integral equations in the infinite stratified domain Ω_o [1]. In particular $\mathbf{u}_o$ may be expressed in terms of a single layer potential on Σ and a Newton potential on Ω_δ [11, 6]. When transient solution are to be obtained, the frequency domain solution is multiplied by a given Fourier spectrum and an inverse Fast Fourier Transform is applied.

3 Variability of the incident field

In this section one assumes that $\Omega_\delta = \emptyset$, Ω_b is bounded and perfectly known and that the incident field is a random process. As the solution $\mathbf{u}$ is a linear function of the incident field, using the linear filtering theory [10] one has only to characterize this input random process and the transfer function relating this input and the desired output : the total field at some given points in the structure. The input of our model - equations (2) and (4) with $\Omega_\delta = \emptyset$ - is the trace of $\mathbf{u}^{inc}$ on Σ . It is characterized as a non-homogeneous non stationary stochastic process derived from the homogeneous stationary process $\tilde{\mathbf{u}}^{inc}(\mathbf{x}, t)$ as follows :

$$(5) \quad \mathbf{u}^{inc}(\mathbf{x}, t) = \varphi(t) \tilde{\mathbf{u}}^{inc}|_\Sigma$$

where $\varphi(t)$ is a deterministic modulation function and $|_\Sigma$ is the restriction on Σ . Moreover as Σ is bounded one knows that $\mathbf{R}^{inc}(\mathbf{x}, \mathbf{x}', \omega)$, the spectral covariance of $\tilde{\mathbf{u}}^{inc}$ has the

following representation :

$$(6) \quad \mathbf{R}^{inc}(\mathbf{x}, \mathbf{x}', \omega) = \sum_{n \in \mathbb{N}^*} \lambda_n \tilde{\mathbf{u}}_n^{inc}(\mathbf{x}) \otimes \tilde{\mathbf{u}}_n^{inc}(\mathbf{x}')$$

where $\tilde{\mathbf{u}}_n^{inc}$ are the eigenmodes of the integral operator of kernel $\mathbf{R}^{inc}$ and λ_n are the corresponding eigenvalues. Then computing the deterministic solution $\tilde{\mathbf{u}}_n$ of the original problem for $\mathbf{u}^{inc} = \tilde{\mathbf{u}}_n^{inc}$ ($\tilde{\mathbf{u}}_n^{inc}$ is the mean of $\tilde{\mathbf{u}}^{inc}$) one can simply express the output as a linear function of these deterministic solutions. As a particular post-processing, the evolutive standard deviation at a given point of the structure reads :

$$(7) \quad \sigma(t) = \int_{\mathbb{R}} \sum_{n \in \mathbb{N}^*} \lambda_n \mathbf{u}_n(t, \omega) \cdot \mathbf{u}_n(t, \omega) d\omega \quad , \quad \mathbf{u}_n(t, \omega) = \mathcal{F}(\tilde{\mathbf{u}}_n \delta_\omega * \hat{\varphi})$$

where $(\delta_\omega * \hat{\varphi})(\omega') = \hat{\varphi}(\omega' - \omega)$, $\hat{\varphi}$ being the Fourier transform of φ and $\mathcal{F}^{-1}$ denoting the inverse Fourier transform. $\sigma(t)$ provides the engineer with bounds on the amplitudes reach by the displacements during the earthquake (see figure 2).

4 Variability of soil parameters

In this section $\mathbf{u}^{inc}$ is assumed to be determinist and Ω_δ bounded. The perturbation $\mathbf{p}_\delta$ is modeled as a centered second order Gaussian stochastic process on Ω_δ characterized by its covariance $\mathbf{R}_p$ and can be approximated as a finite Kahrnen-Loeve expansion [7] :

$$(8) \quad \mathbf{p}_\delta = \sum_{n=1}^N \theta_n \mathbf{p}_n$$

where $\mathbf{p}_n / \sqrt{\lambda_n}$ and λ_n are the first eigenmodes and eigenvalues of the covariance operator associated to the kernel $\mathbf{R}_p(\mathbf{x}, \mathbf{x}')$ (see section 3 or [7]) and $\boldsymbol{\theta} = \{\theta_n\}_{n=1, N}$ is a vector of normalized centered Gaussian uncorrelated random variables on $\mathbb{R}$. The associated parameter space denoted H_p is then a finite dimension Gaussian Hilbert space for the classical weighted scalar product based on the joint probability functions $dp_n(\theta_n, \theta'_n)$. Then the displacement field can be modeled as a stochastic process $\mathbf{u}(\mathbf{x}, \boldsymbol{\theta})$ belonging to the Gaussian Hilbert space $L^2(\Omega_p, V)$ for the scalar product $E \langle \cdot, \cdot \rangle$, $E \langle \cdot \rangle$ denoting the mathematical expectation (see [8, 9]).

At last, using the deterministic parameters fields $\mathbf{p}_n$ on Ω_δ to built the associated deterministic bilinear forms $k_n(\cdot, \cdot)$ satisfying $k_\delta = \sum_n \theta_n k_n$ and taking the mathematical expectation of the original deterministic problem for any $\mathbf{v}$ in $L^2(\Omega_p, V)$ one gets the following equation :

$$(9) \quad E \langle (\mathbf{f}_s, \mathbf{v}_\delta)_\delta \rangle = E \langle k_o(\mathbf{u}_o, \mathbf{v}_o) \rangle + \sum_n E \langle \theta_n k_n(\mathbf{u}_\delta, \mathbf{v}_\delta) \rangle + E \langle k_b(\mathbf{u}_b, \mathbf{v}_b) \rangle$$

approximating $\mathbf{u}(\mathbf{x}, \boldsymbol{\theta})$ and $\mathbf{v}(\mathbf{x}, \boldsymbol{\theta})$ on a finite basis $\mathbf{w}_k(\mathbf{x}) \psi_l(\boldsymbol{\theta})$ leads to an huge linear system to solve in the unknown coefficients u_{kl} , k indicating the spatial approximation and l the parametric one. Usually basis functions $\mathbf{w}_k(\mathbf{x})$ are the same as for deterministic analyses whereas $\psi_l(\boldsymbol{\theta})$ are either the Hermite polynomials for spectral methods or piecewise constant for Monte Carlo simulations. In the first case the system is coupled both regarding the space and the parameter variables and in the second one it is uncoupled regarding the parameter variables leading to smaller independent linear systems to solve. Even though parallel

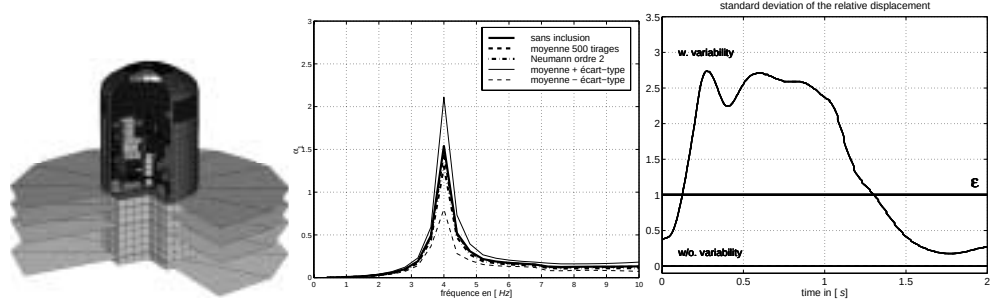


FIG. 2. Influence of the soil variability on the seismic response of a reactor building ($r = 25m$) : (left) the mesh, (center) the frequency response on top for a standard deviation of 80% and a correlation length of 50m, (right) Evolutive standard deviation of the relative displacements between two sub-structures compared to the tolerance ϵ .

computing may significantly reduce the computational effort, a large number of simulations have to be performed to span the parameter space. Factorizing once the deterministic part of equation (9) using Integral Operators [11, 6] and using variance reduction techniques allow the computation of realistic models such as a reactor Building resting on a stratified half-space perturbed by large random fluctuation under the foundation (figure 2).

Once the approximation of $\mathbf{u}(\mathbf{x}, \boldsymbol{\theta})$ computed, the statistical properties of the stochastic process $\mathbf{U}(\mathbf{x}) = \{\mathbf{u}(\mathbf{x}, \boldsymbol{\theta}), \boldsymbol{\theta} \in \Omega_p\}$ can be estimated. For example at a given nodal point $\mathbf{x}_I$, $\bar{U}_{Ii}$, the mean value of the displacements in direction $\mathbf{e}_i$ and C_{Ii} , the associated covariance in the same direction reads, assuming that $\mathbf{w}_k$ is the FE basis function associated to this node and this direction :

$$\bar{U}_{Ii} = E \langle \mathbf{u}(\mathbf{x}, \boldsymbol{\theta}) \rangle = \sum_l u_{kl} E \langle \psi_l(\boldsymbol{\theta}) \rangle, \quad C_{Ii}^2 = \sum_l |u_{kl}|^2 E \langle \psi_l(\boldsymbol{\theta})^2 \rangle$$

5 The city effect

In the section the soil and the incident field are assumed to be deterministic and one concentrates on the randomness of the building distribution. Then as long as Ω_b remains finite, the methods presented in section 4 to account for the soil variability apply. However in order to represent an entire city, some additional work has to be done. The basic idea here is two folds : i) there is no correlation between two buildings, ii) the building distribution is homogeneous on P_o . In such situations it is usually assumed that periodic models are able to represent efficiently this random distribution as long as the size of the periodic cells are much greater than the correlation length and the mean-path between heterogeneities. One can then apply a Floquet Transform to the field equations and restricts the analysis to a generic cell. On this cell using integral equations based on Green functions satisfying the periodicity conditions one do not have to care about the infinite boundaries of the cell (see [4, 5]). Then one is led back to sections 3 or 4 where Σ and Ω_δ were bounded and the same techniques apply.

For our practical application the random building distribution is generated by an heuristic method (One throws storey randomly inside a given block) satisfying in a mean sense architectural requirements (one stops when the allowed built surface is reached) and structural engineering ones (The fundamental natural period of a building is a linear function of the number of storeys). Once the numerical solution computed, statistical

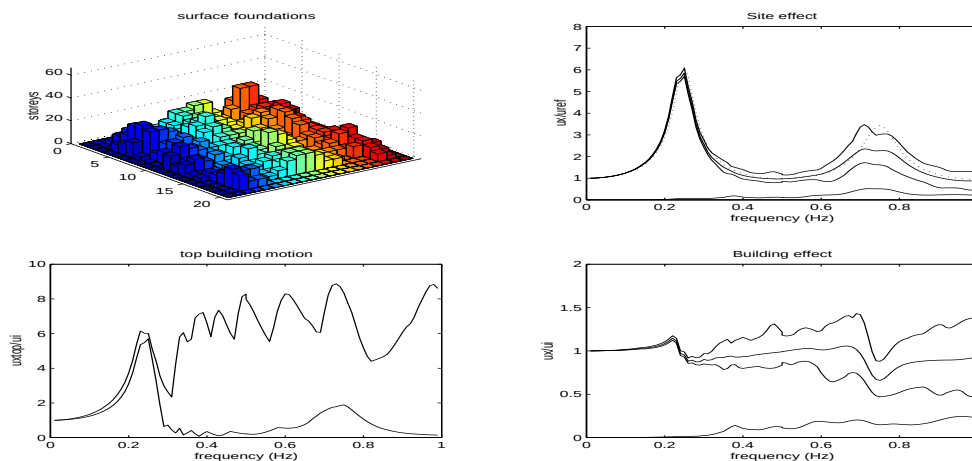


FIG. 3. Diffusion of the vertical incident wave on a random building distribution resting on soft layer : (top-left) the random distribution, (top-right) min.-mean-max.-cov. of the site response at the basement of the buildings, (bottom-left) mean-max. of the building response at the top, (bottom-right) min.-mean-max.-std. amplifications due to buildings

analysis are performed on the seismic field showing in particular strong diffusion effects (see figure 5).

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